5	(a)	Electric field strength at a point in an electric field is defined as the electric force per unit charge on a positive charge placed at that point. [1]	B1
	(b)(i)	$E = V/d = 4.5 / (0.030)$ $= 150 \text{ V m}^{-1}$	C1 A1
	(b)(ii)	Loss in KE = gain in EPE or $KE_B + EPE_B = KE_A + EPE_A$ $\frac{1}{2} m u^2 = e V$ or $\frac{1}{2} m u^2 + (-1.6 \times 10^{-19})(0) = 0 + (-1.6 \times 10^{-19})(-4.5)$ $u = \sqrt{\frac{2(1.6\times10^{-19})(4.5)}{9.11\times10^{-31}}}$ [1] $= 1.26 \times 10^6 \text{ m s}^{-1}$ [1] Alternative (Kinematics): $a = eE/m$ $v^2 = u^2 + 2as$ $0^2 = u^2 + 2\left[\frac{(-1.6 \times 10^{-19})(150)}{9.11 \times 10^{-31}}\right](30 \times 10^{-3})$ $u = 1.26 \times 10^6 \text{ m s}^{-1}$	C1 A1
	(b)(iii)	The minimum value of u in order for electrons to reach plate A is dependent on the potential difference between the plates. [1] Thus the value in (b)(ii) is unchanged. [1] If student states that it is independent of distance and that the value remains unchanged, he will get 1 mark for the A1.	M1 A1
	(c)(i)	$Q = CV = (160 \times 10^{-6})(200) = 0.032 \text{ C}$ [1]	A1
	(c)(ii)	$\Delta E = \frac{1}{2} C (V_2^2 - V_1^2) = \frac{1}{2} (160 \times 10^{-6})(200^2 - 100^2)$ [1] $= 2.4 \text{ J}$ [1]	C1 A1